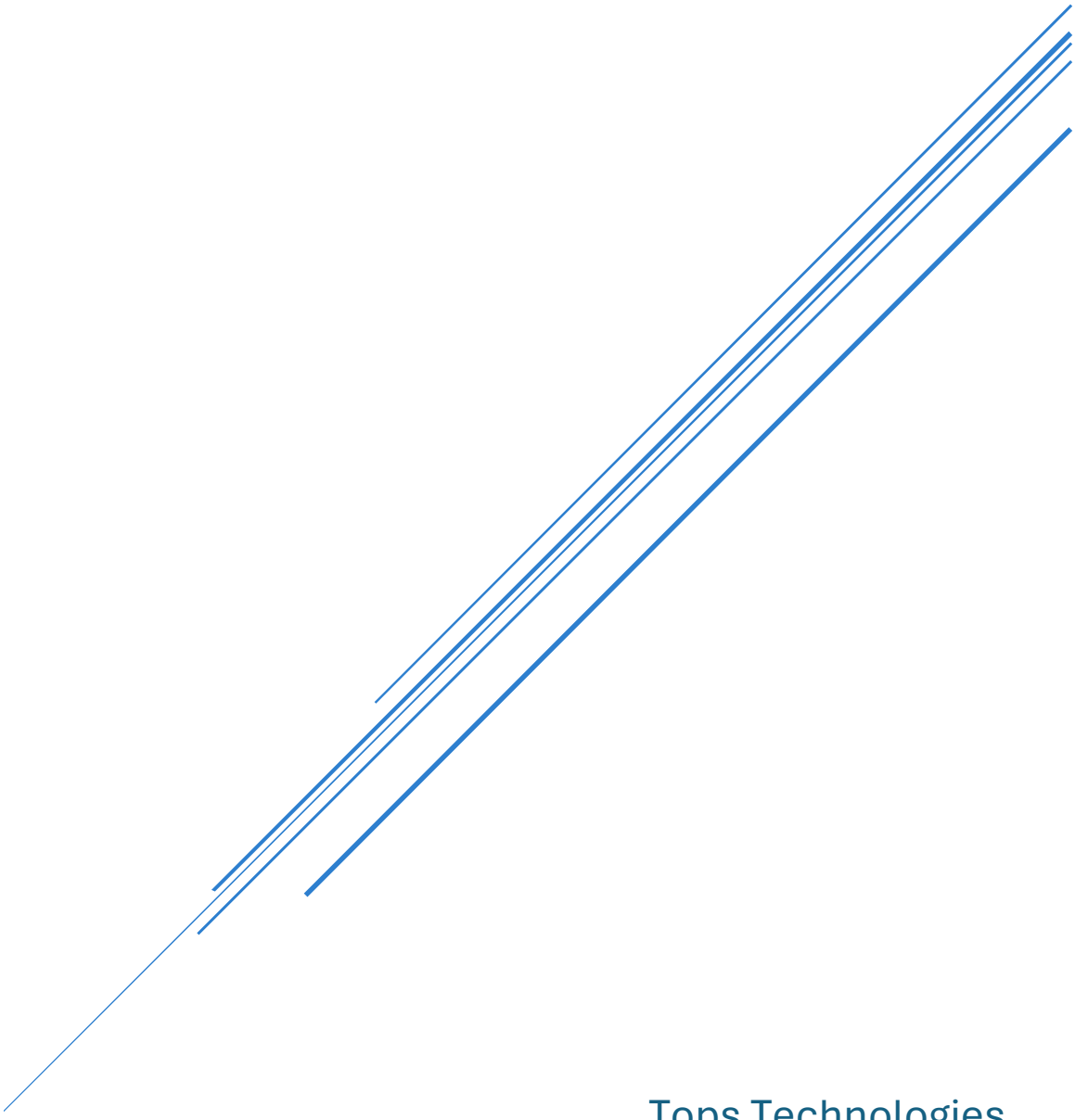
SWITCHING AND VLANS

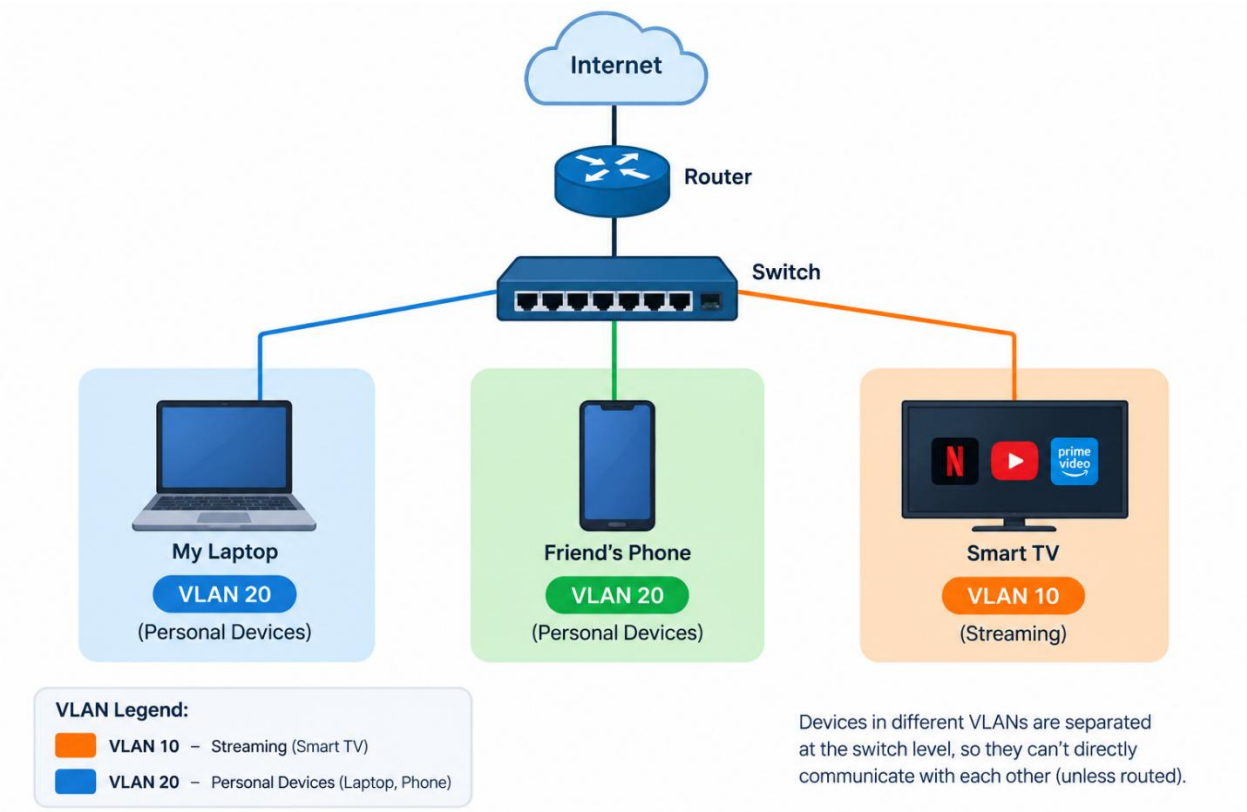
Assignment - 10



Tops Technologies
Cyber Security

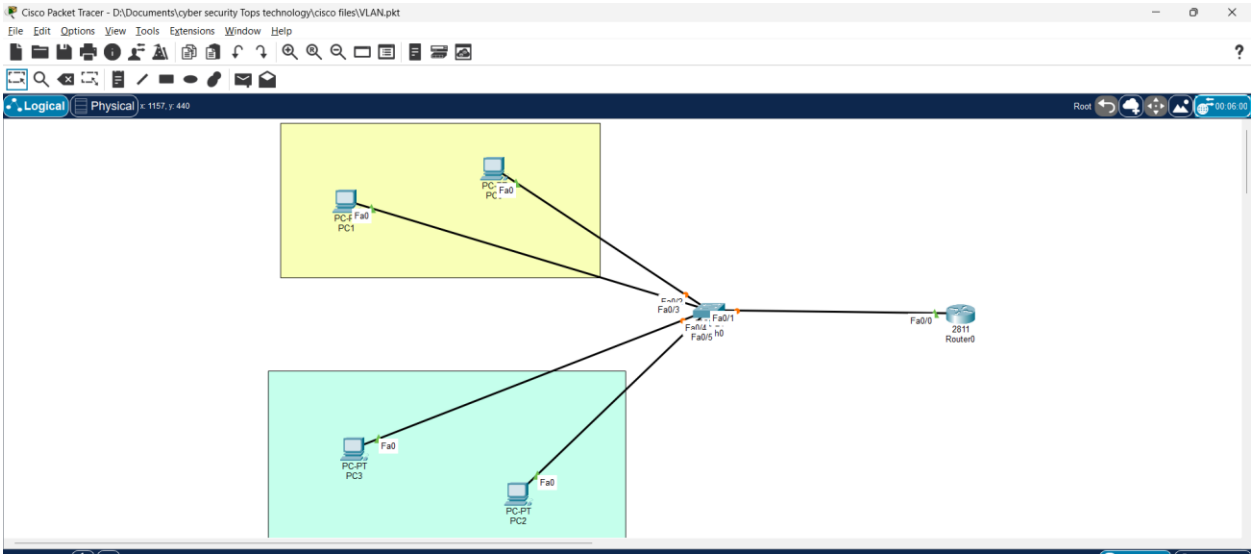
1. **Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).**

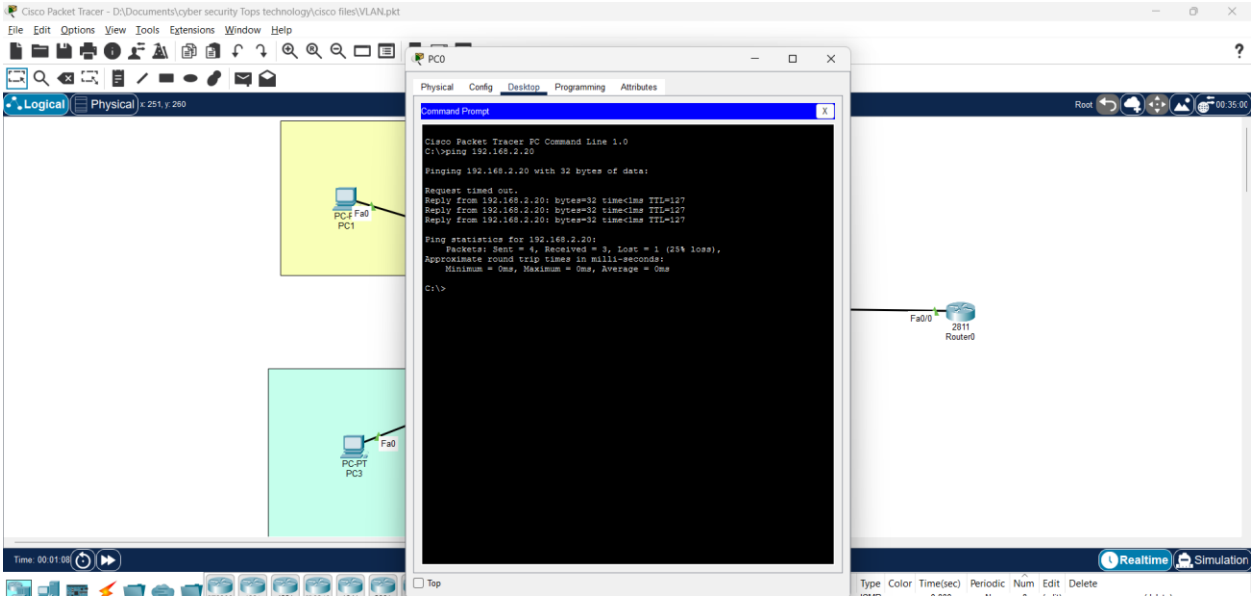
➢



2. **Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.**
   Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

➢

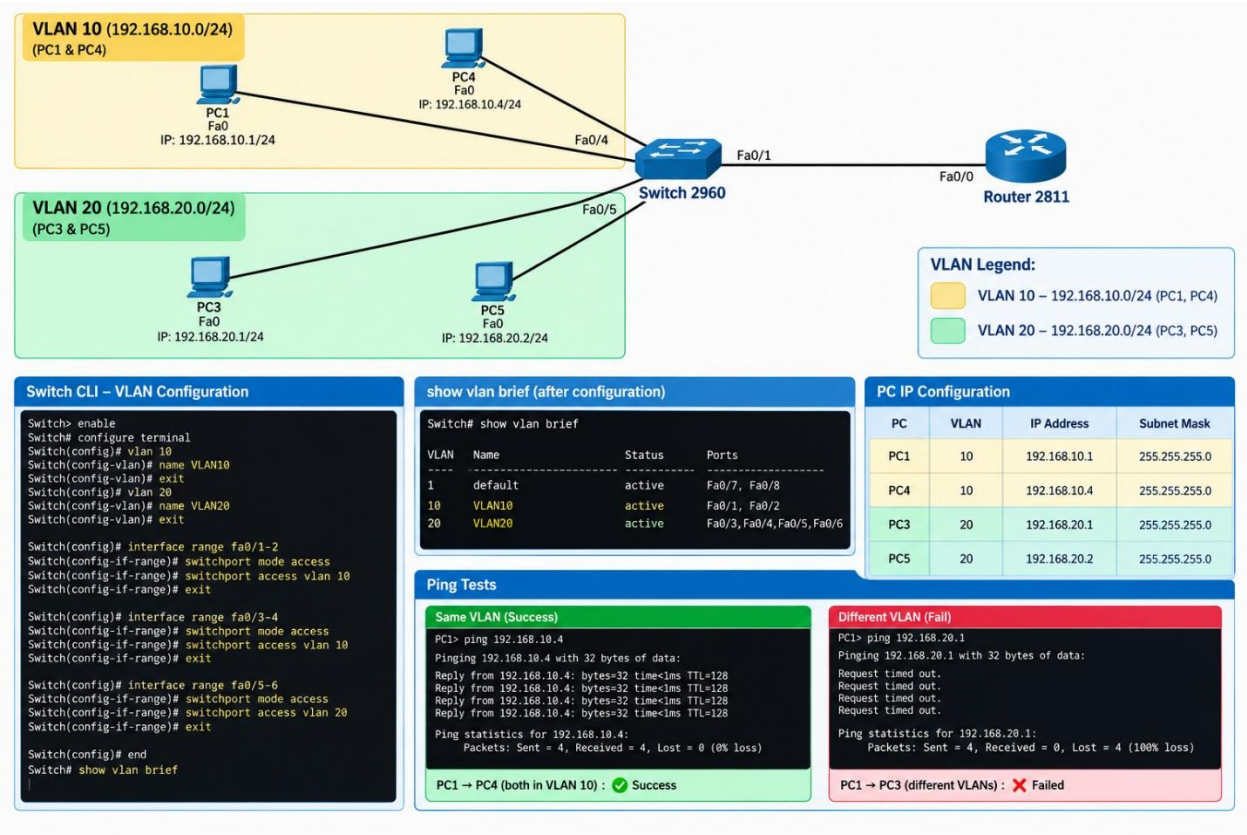




The switch was configured with two VLANs: VLAN 10 and VLAN 20. Three PCs were assigned to VLAN 10, while two PCs were assigned to VLAN 20. The show vlan brief command was used to verify the VLAN membership. Ping tests showed that PCs in the same VLAN could communicate, whereas PCs in different VLANs could not communicate without inter-VLAN routing.

3. **In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.**

➢



4. **Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).**
   Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

➢

1. Switch CLI commands

Switch> enable

Switch# configure terminal

Switch(config)# vlan 30

Switch(config-vlan)# name VLAN30

Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet 0/1-10

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 30

Switch(config-if-range)# end

Switch# show vlan brief

2. Expected result

The show vlan brief command should show VLAN 30 as active, with ports Fa0/1 through Fa0/10 assigned to it.

3. Explanation

The vlan 30 command creates VLAN 30. The interface range fastEthernet 0/1-10 command selects all 10 ports at once. The switchport access vlan 30 command assigns those ports to VLAN 30 without repeating the command for each port.